MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5840W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Pentode, subminiature, sharp cutoff

Outline	---	3-1 (EIA)
Base	---	E8-10
Envelope	---	T3.
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	g1	k,g3	h	k,g3	a	h	g2	k,g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rk	Rg1	IK	Pp	Pg2	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	v	Ohms	Meg	mAdc	W	W	°C	ft
Maximum:	6.6	165	0, -55	155	200	---	1.1	16.5	1.1	0.55	+220	(See note 1)
Minimum:	6.0	---	---	---	---	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	100	0	100	0	150	---	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)			0.4	II	Ib	5.5	9.5	mAdc
1256	Electrode current (2) (anode)		Ec1 = -9.0 Vdc; Rk = 0	0.4	II	Ib	---	50	µAdc
1256	Electrode current (screen grid)			0.4	II	Ic2	1.5	3.3	mAdc
1266	Total grid current	6	Rg1 = 1.0 Meg	0.4	II	Ic	0	-0.3	µAdc
1301	Heater current			0.4	II	If	140	160	mA
1306	Transconductance (1)			0.4	II	Sm	4,200	5,800	µmhos
1336	Heat-cathode leakage			0.4	II	Ihk	---	5.0	µAdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	I	---	---	---	---
1031	High-frequency vibration		Rp =10,000 ohms	6.5	Code H	Ep	---	60	mVac
1246	Audio frequency noise	3	Esig = 70 mVac; Ec2 = 19 Vdc; Rg1 = 0.1 Meg; Rg2 = 1,000 ohms; Rp = 0.2 Meg	2.5	I	---	---	---	---
1266	Grid emission	2	Ef = 7.5 V; Ec1 = -9.0 Vdc; Rg1 = 1.0 Meg; Rk = 0	2.5	I	Ic1	0	-0.5	µAdc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	10	%
1311	Anode resistance			2.5	I	rp	0.175	---	Meg
1331	Direct-interelectrode capacitance		0.405 in. dia shield	6.5	Code E	Cg1p Cin Cout	--- 3.5 2.9	0.015 4.9 3.9	pF pF pF
1116	Lead fatigue			2.5	Code F	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1041	Shock	4	450 G; Ehk = +100 Vdc	---	See note 5	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
1031	Vibration-fatigue			6.5	See note 5	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	High-frequency vibration		Rp = 10,000 ohms	---	---	Ep	---	200	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	20	µAdc
1306	Change in trans-conductance (1) of individual tubes			---	---	ΔSmt	---	20	%
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.0 V; 1 min "on", 4 min "off"; Ehk = 140 Vac; Ec1 = Ec2 = Eb = 0	---	---	---	---
1516	Stability life		Ehk = +200 Vdc; Rg = 1.0 Meg; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	10	%
1501	Intermittent life (room temperature)	7-9	Ehk = +200 Vdc; Rg = 1.0 Meg	---	---	---	---
---	Intermittent life (room temperature) - test end point (1,000 hours):						
1211	Insulation of electrodes			R	25	---	Meg
1501	Intermittent life (high temperature)	7-9	Ehk = +200 Vdc; Rg = 1.0 Meg; TE = +220°C (min)	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3 -Continued						
---	Intermittent life (high temperature) - test end points (1,000 hours):						
1266	Total grid current		Rg1 = 1.0 Meg	Ic	0	-0.8	µAdc
1301	Heater current			If	138	164	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	25	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSm Ef	---	20	%
1336	Heater-cathode leakage			Ihk	---	10	µAdc
1211	Insulation of electrodes			R	25	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

2. Prior to this test, the tube under test (TUT) shall be preheated a minimum of 5 minutes at the conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Ec1	Ec2	Ec3	Eb	Rk	Rg1
V	Vdc	Vdc	Vdc	Vdc	Ohms	Meg
7.5	0	100	0	100	150	1.0

3. The rejection level shall be set at the VU meter reading obtained during calibration.

4. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.

5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

6. This test shall be performed at the conclusion of the holding period.

7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

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8. Eligibility for early release at 500 hours is established if no lot failure due to this life test or the 1,000-hour intermittent (high temperature) life test has occurred in the preceding three consecutive lots and the lot under consideration is acceptable at 500 hours. One lot failing the 1,000-hour intermittent (high temperature) life test or the 1,000-hour intermittent (room temperature) life test, shall result in loss of eligibility for the early release procedure. All intermittent (room temperature) life-test samples shall be operated to 1,000 hours.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG, SH
Air Force - 11

Agent:
DLA - ES

Preparing activity:
Navy - EC

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